Preliminary.

$(\mathbb{Z}_n, +, \cdot)$ is a Ring. If $n=p$ is prime, then for any $a \in \mathbb{Z}_p$, because

$$a^{p-1} \equiv 1 \mod p.$$

That is, $a^{-1} = a^{p-2} \mod p \in \mathbb{Z}_p$.

Let F be a field. If $ch(F)=0$, then F must contains $\mathbb{Q}$,
If $ch(F) \neq 0$, then F must contains $\mathbb{F}_p$, where $p = ch(F)$

If $\deg_{\mathbb{F}_p} \alpha = n$, then $F(\alpha)$ has p^n elements.

Because $F(\alpha) \cong \mathbb{F}_p^n$

If E is finite field, then E has p^n elements for some $n \in \mathbb{N}$, where $p = ch(E)$.
Because since E is finite, $\dim_{\mathbb{F}_p} E$ must be finite.

Thm. Let $\mathbb{Z}_p \leq E \leq \overline{\mathbb{Z}_p}$ has p^n elements. Then, for $p(x) = x^{p^n} - x \in \mathbb{Z}[x]$,

$$E = \{\alpha \in \overline{\mathbb{Z}_p} \mid p(\alpha) = 0\}$$

Proof. Since $(E \setminus \{0\}, \times)$ is a group of order $p^n - 1$, for $\alpha \in E \setminus \{0\}$, $|\alpha|$ divides $p^n - 1$.
So that $\alpha^{p^n - 1} = 1$, therefore $p(\alpha) = 0$. And, the polynomial $p(x)$ can have at most p^n zeros,
Consequently we obtain the result.

Thm. Let F be a finite field of characteristic p.
Then, a map

$$\varphi_p : F \to F : x \mapsto x^p$$

is Automorphism. Moreover, $F_{\{\varphi_p\}} = \mathbb{Z}_p$.

Proof. Well-defined is clear.

Homomorphism: for any $a, b \in F$,

$$\varphi_p(a+b) = (a+b)^p \underset{\text{Binomial}}{=} \sum_{k=0}^{p} \binom{p}{k} a^{p-k} b^k \underset{ch(F)=p}{=} a^p + b^p = \varphi_p(a) + \varphi_p(b)$$

And clearly preserves multiply.

Injectivity:

$$\varphi_p(a) = a^p = 0 \iff a = 0 \quad \Rightarrow \ker \varphi_p = \{0\}.$$

Surjectivity: Finite injective endomorphism, thus surjective.

Meanwhile, by the Fermat's Theorem, for any $a \in \mathbb{Z}_p$, (Isomorphically the prime field of F)

$$\varphi_p(a) = a^p = a$$

Thus $\mathbb{Z}_p \subseteq F$. And, observe that

$$\varphi_p(x) = x \iff x^p = x \iff x^p - x = 0.$$

This polynomial has at most p zeros in F, thus $\mathbb{Z}_p = F_{\{\sigma_p\}}$.

Corollary. If $ch(F) = p$, then $(\alpha+\beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$ for any $\alpha, \beta \in F$, $n \geq 1$.

「The existence of $GF(p^n)$.

「Thm. If F is a field with $ch(F) = p$, then $x^{p^n} - x \in F[x]$ has p^n distinct zeros in $\overline{F}$.

Proof. observe that $x^{p^n} - x = x \cdot [x^{p^n-1} - 1]$. Let $f(x) = x^{p^n-1} - 1$.

Suppose that $\alpha \in \overline{F}$ is a root of $f(x)$. Then

$$f(x) = (x-\alpha)\cdot\underbrace{(x^{p^n-2} + \alpha x^{p^n-3} + \cdots + \alpha^{p^n-3}x + \alpha^{p^n-2})}_{p^n-1 \text{ terms}}$$

Let $g(x) = f(x)/(x-\alpha)$. Now,

$$g(\alpha) = [(p^n-1)\cdot 1]\cdot\alpha^{-1} = -\alpha^{-1} \quad (\text{since } (p\cdot 1)x = 0,\ ch(F) = p)$$

But $\alpha \neq 0$, thus $-\alpha^{-1} \neq 0$. This means α is a zero of f of multiplicity 1.

「A finite field of order of p^n exists.

Proof. Let $K = \{x \in \overline{\mathbb{Z}_p} \mid x^{p^n} - x = 0\} \subseteq \overline{\mathbb{Z}_p}$